

Karthikheyaa Kurra

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EDUCATION

University of Delaware

M.S. Data Science, GPA 3.80/4.00

Newark, DE

Aug 2024 – May 2026

- **Coursework:** Probability Theory & Applications, Mathematical Techniques for Data Science (Optimization), Applied Multivariate Statistics, Machine Learning, Algorithm Design & Analysis, Research Software Engineering

Sathyabama Institute of Science & Technology

B.E. Computer Science & Engineering

Chennai, India

Oct 2020 – May 2024

EXPERIENCE

Research Associate, Scientific Machine Learning

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Jun 2026 – Present

Newark, DE

- Authoring a paper on numerical Partial Differential Equation solvers that combine classical methods and neural networks to solve a nonlinear elliptic PDE on a large domain without linearizing the equation
- Built a Newton solver that generates synthetic training data by solving nonlinear PDE problems using chebyshev discretization to machine precision and produces reliable ground truth with quadratic error decay
- Trained a deep learning model to predict the solution field with a relative error of 0.8% on the high frequency variable co-efficient PDE test data
- Developed and modularized a reproducible and reusable research codebase using Git for version control to maintain the experiment tracking and comprehensive documentation

Software Engineer Intern

Best Egg, University of Delaware (Advisor: Mike Urban)

Jan 2026 – May 2026

Newark, DE

- Contributed to a software system to process financial documents (bank statements, paystubs, tax), by encrypting personal information, and extracted income and balance data for bank review
- Built statistical and AI models to generate confidence scores with a 95% threshold to predict the loan approval for the customers
- Enforced safety protocols and guardrails to detect patterns for fraudulent documents and to handle edge cases like incorrect or insufficient documents
- Deployed the system as 3 services on Google Cloud Run with CI/CD through GitHub Actions, and set up a Grafana dashboard to track elements like throughput, failures, and redaction counts

Research Assistant, Scientific Machine Learning

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Jun 2025 – Jun 2026

Newark, DE

- Implemented a hierarchical linear partial differential equation (PDE) solver using spectral methods and Chebyshev discretization to efficiently compute large domains.
- Validated the solver using the method of manufactured solutions across 6 PDE fields, achieving $O(N \log N)$ time complexity and near-zero spectral accuracy (machine precision).

Research Intern, Computer Vision

Indian Institute of Technology, Hyderabad

Mar 2023 – Jun 2023

Hyderabad, India

- Case study applying supervised contrastive learning (Khosla et al., 2020) to facial expression recognition, comparing SupCon loss against a standard cross-entropy baseline in PyTorch with face detection and data augmentation.
- Analyzed learned t-SNE embeddings with models producing accuracy of 80.7% and 79.4% respectively on imbalanced data

Teaching Assistant, Machine Learning

Sathyabama Institute of Science & Technology

Jan 2023 – Apr 2023

Chennai, India

- Graded assignments, homework; prepared lecture slides, supplementary notes for the undergraduate machine learning course
- Led review sessions on mathematics for machine learning, covering matrix algebra and multivariable calculus.

Summer Intern

The University of Texas at Dallas

Jun 2022 – Jul 2022

Dallas, TX

- Intensive summer program in machine learning and deep reinforcement learning: lectures, labs, and hands-on model building from first principles.

TECHNICAL SKILLS

Languages: Python, MATLAB, R, SQL, Bash, LaTeX

Scientific ML: PyTorch, NumPy/SciPy, Operator Learning, Physics-Informed Neural Networks (PINNs)

Tools: Git, Docker, Linux, HPC job scripting, VS Code, Cursor

READING GROUPS

Scientific Machine Learning Reading Group

University of Delaware (Advisor: Dr. Ke Chen)

2025 – Present

Newark, DE

- Weekly seminar with rotating student presentations of research literature; topics include Computational Optimal Transport (Peyré & Cuturi), Equivariant Neural Networks, and One-Shot Learning.